© 2017 Nature America, Inc., part of Springer Nature. All rights reserved.

Accessible, curated metagenomic data through ExperimentHub

To the Editor: The microbiome has emerged as a key aspect of human biology and has been implicated in many disease etiologies. Shotgun metagenomic sequencing is an approach with the highest resolution currently available for studying the taxonomic composition and functional potential of the human microbiome. The increase in publicly available shotgun data theoretically enables hypothesis testing for specific diseases and environmental niches as well as meta-analysis across related studies. However, several factors prevent the research community from taking full advantage of these resources. Barriers include the need for substantial investments of time, computational resources and specialized bioinformatic expertise as well as inconsistencies in annotation and formatting between individual studies.

To overcome these challenges, we developed the curatedMetagenomicData data package (https://waldronlab.github.io/curatedMetagenomicData/ and **Supplementary Software**) for distribution through the Bioconductor[1] ExperimentHub platform (**Supplementary Methods**). curatedMetagenomicData provides uniformly processed human microbiome data including bacterial, fungal, archaeal, and viral taxonomic abundances, in addition to quantitative metabolic functional profiles and standardized per-participant metadata. Data resources are accessible to users with minimal bioinformatic knowledge; while integration with the R/Bioconductor environment allows flexibility for biologists, clinicians, epidemiologists, or statisticians to perform novel analyses and methodological development, and integration of resources. We produced these resources by (i) downloading raw sequencing data; (ii) processing the data through the MetaPhlAn2 (ref. 2) and HUMAnN2 (ref. 3) pipelines; (iii) manually curating sample and study information (**Supplementary Table 1**); (iv) representing the above results as documented, integrative Bioconductor objects; and (v) working with the Bioconductor core team to develop the ExperimentHub platform for efficient distribution (**Fig. 1**). ExperimentHub is a novel platform for scalable distribution of experimental data to the R desktop. It allows distributers and downloaders to access metadata through a Bioconductor-hosted SQL database and to access bulk data through Amazon S3 buckets via a standard R software package. An extension permits command-line access. The current version of curatedMetagenomicData (v1.7.5) includes samples from multiple body sites profiled by the Human Microbiome Project[4] and from 25 other metagenomic studies. These total 5,716 samples that span 34 diseases and 28 countries (**Supplementary Table 2**).

Using curatedMetagenomicData and the statistical, visualization, and microbial ecology tools available in R/Bioconductor makes many analyses much simpler and more powerful than previous analyses. As a demonstration, we employed a random-forest algorithm on three taxonomic data types (species abundance, genetic marker presence, and abundance) and two functional abundance profiles (pathway abundance and coverage) to develop predictive models of diabetes, inflammatory bowel disease, cirrhosis, colorectal cancer, and obesity. Cross-validation prediction accuracy varied substantially for these different outcomes, but the five data types provided nearly identical prediction accuracy (**Fig. 1**, example 1). Second, we performed unsupervised clustering of human gut microbiome profiles. In a large combined data set (n = 3,667), we observed that micro-

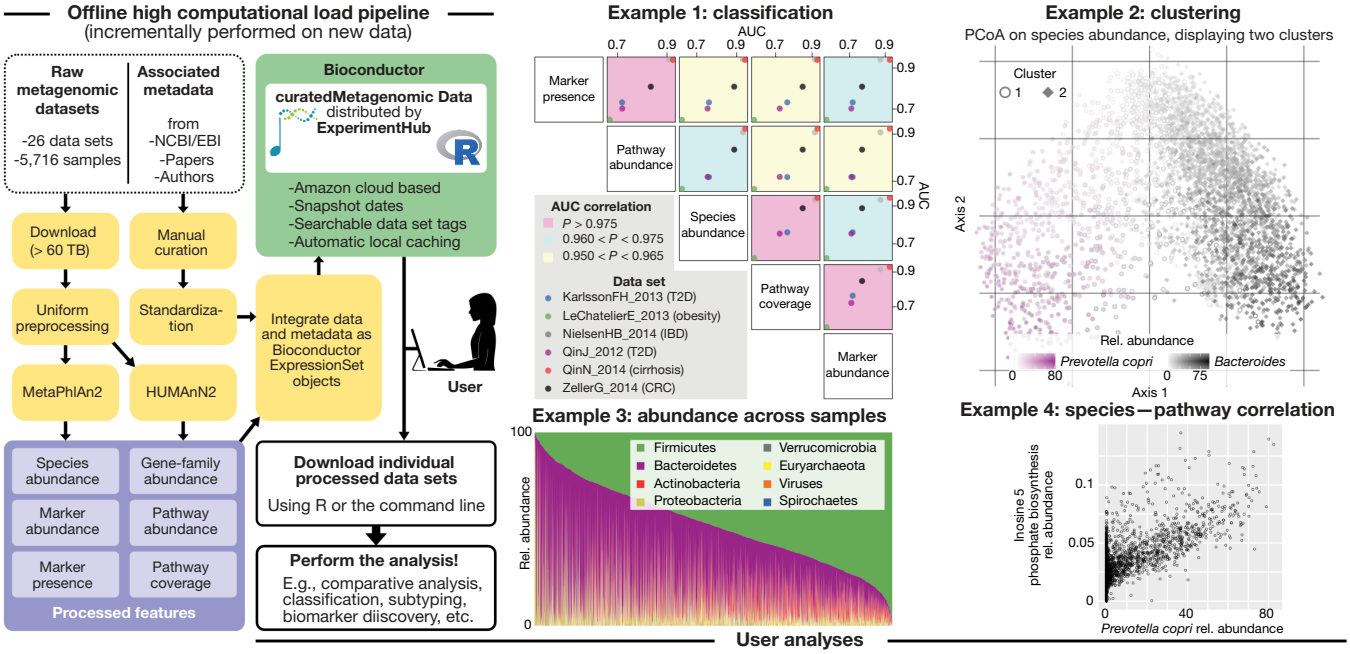

Figure 1 | curatedMetagenomicData production pipeline and examples of enabled analyses. The pipeline (left) processes raw metagenomic sequence data to produce taxonomic and functional profiles, integrates these with curated sample data, then documents and packages these for distribution through ExperimentHub as the curatedMetagenomicData package. Example 1: health status was classified for six data sets, five data types each, using a random-forest algorithm and cross-validation to estimate prediction accuracy. Example 2: unsupervised clustering of human gut samples (n = 3,667) using the Bray–Curtis metric and partitioning around medoids. In Cluster 1, *Bacteroides* relative abundance is shown by point borders and *Prevotella copri* by centers. Example 3: abundance at the phylum level in human gut samples. Example 4: correlation between *Prevotella copri* and inosine 5 phosphate biosynthesis, the most correlated species–pathway pair (see **Supplementary Fig. 4** for heatmap of most highly correlated species–pathway pairs). These analyses were performed using the script provided in the vignettes/extras package directory.

bial communities are strongly patterned by a continuous gradient between relative abundance of *Prevotella copri* and *Bacteroides spp* (**Fig. 1**, example 2) with little or no support for any discrete clusters (**Supplementary Fig. 1**). This is consistent with the analysis of Koren *et al.*[5] but stands in contrast with the three-enterotypes hypothesis of Arumugam *et al.*[6]. Third, we visualized the continuum of the firmicutes–bacteroidetes gradient in gut microbiomes (**Fig. 1**, example 3). This continuum has been previously reported[4], but not for thousands of microbial species on thousands of individuals as analyzed here. Finally, we ranked all taxa–pathway pairs by magnitude of correlation in samples. The highest correlation pair shown demonstrates a strong relationship between *Prevotella copri* abundance and inosine 5 phosphate biosynthesis (**Fig. 1**, example 4), and this suggests functional differences along the gradient shown in example 2. These and other analyses (**Supplementary Figs. 1–5**) would be very large undertakings using less curated databases such as IMG/M or EBI Metagenomics; but they are straightforward, documented, and reproducible using curatedMetagenomicData.

Our large-scale, curated integration of metagenomic data is well documented and readily usable by broad scientific communities for efficient hypothesis testing and methods development. The automated pipeline allows the resource to be continually expanded by the team and community contributors (**Supplementary Methods**, package maintenance). By allowing researchers to bring their expertise to the analysis of metagenomic data without the need for extensive bioinformatic experience, curatedMetagenomicData greatly expands the accessibility of public data for study of the human microbiome.

Data availability statement. The curatedMetagenomicData data package is available online at https://waldronlab.github.io/curatedMetagenomicData/ and through the Bioconductor package installer.

A **Life Sciences Reporting Summary** is available.

Note: Any Supplementary Information and Source Data files are available in the online version of the paper (doi:10.1038/nmeth.4468).

ACKNOWLEDGMENTS
This work was made possible by the CUNY High Performance Computing Center, College of Staten Island, funded in part by the City and State of New York, CUNY Research Foundation, and National Science Foundation Grants CNS-0958379, CNS-0855217 and ACI 1126113. Support was provided by the European Union H2020 Marie-curie grant (707345) to E.P., the European Research Council (ERC-STG project MetaPG), MIUR "Futuro in Ricerca" RBFR13EWWI_001, the People Programme (Marie Curie Actions) of the European Union Seventh Framework Programme (FP7/2007-2013) under REA grant agreement no. PCIG13-GA-2013-618833, the LEO Pharma Foundation, and by Fondazione CARITRO fellowship Rif.Int.2013.0239 to N.S., the National Institute of Allergy and Infectious Diseases (1R21AI121784-01 to J.B.D. and L.W.) and the US National Cancer Institute (U24CA180996 to M.M. and L.W.).

COMPETING FINANCIAL INTERESTS
The authors declare no competing financial interests.

Edoardo Pasolli[1,8], Lucas Schiffer[2,3,8], Paolo Manghi[1,8], Audrey Renson[2,3], Valerie Obenchain[3], Duy Tin Truong[1], Francesco Beghini[1], Faizan Malik[2], Marcel Ramos[2–4], Jennifer B Dowd[2,5], Curtis Huttenhower[6,7], Martin Morgan[4], Nicola Segata[1] & Levi Waldron[2,3]

[1]Centre for Integrative Biology, University of Trento, Trento, Italy. [2]Graduate School of Public Health and Health Policy, City University of New York, New York, New York, USA. [3]Institute for Implementation Science and Population Health, City University of New York, New York, New York, USA. [4]Roswell Park Cancer Institute, University of Buffalo, Buffalo, New York, USA. [5]Department of Global Health and Social Medicine, King's College London, London, UK. [6]Biostatistics Department, Harvard School of Public Health, Boston, Massachusetts, USA. [7]The Broad Institute, Cambridge, Massachusetts, USA. [8]These authors contributed equally to this work. Correspondence should be addressed to L.W. (levi.waldron@sph.cuny.edu) or N. S. (nicola.segata@unitn.it)

1. Huber, W. *et al. Nat. Methods* **12**, 115–121 (2015).
2. Truong, D.T. *et al. Nat. Methods* **12**, 902–903 (2015).
3. Abubucker, S. *et al. PLoS Comput. Biol.* **8**, e1002358 (2012).
4. Human Microbiome Project Consortium *Nature* **486**, 207–214 (2012).
5. Koren, O. *et al. PLoS Comput. Biol.* **9**, e1002863 (2013).
6. Arumugam, M. *et al. Nature* **473**, 174–180 (2011).

The Experimental Design Assistant

To the Editor: The quality and reliability of much animal research is in question. Unreliable or low-quality research represents an unacceptable waste of animals and research resources. In the US alone, the cost of irreproducible research has been estimated at $28 billion annually, and issues with research design and reporting are estimated to account for half of that waste[1]. To address these issues, the National Centre for the Replacement, Refinement and Reduction of Animals in Research (NC3Rs) developed the ARRIVE guidelines to improve the reporting of *in vivo* research[2,3]. We now present the Experimental Design Assistant (EDA; https://eda.nc3rs.org.uk), a freely accessible web-based tool, which was launched to help researchers improve the design, conduct, analysis and reporting of animal experiments.

The system was developed by progressive interaction between an expert group experienced in providing advice on experimental design to researchers and a software development team. It includes a computer-aided design tool through which the user develops a diagram that embodies the experimental plan. The diagram offers a new standard notation for describing experiments in which methodological details and analysis plans are explicit (**Fig. 1**). This facilitates communication between collaborators, funding bodies, ethical review committees, journal editors and peer reviewers; it also allows detailed record keeping and may serve as an *ex ante* registered protocol[4].

The structure of EDA diagrams is based on a series of relationships between the different components of the experiment. This allows the use of computer-based logical reasoning to provide feedback and advice on the experimental plans[5]. The feedback helps researchers improve their experimental design by, for instance, highlighting missing information or problems with internal consistency. It also provides assistance with identifying and characterizing the independent variables and outcome measures to be included in the analysis. Advice about common nuisance variables, which threaten the internal validity of many animal experiments, is also provided, along with practical suggestions to account for such variables in randomization and data analysis. The feedback does not restrict researchers to using a particular design type, but it promotes a better understanding of the implications of common design pitfalls so that researchers can make informed decisions. The feedback also suggests methods of statistical analysis that are appropriate for a given design, along with advice on any data requirements ('assumptions') for a given test as well as possible data transformations.

Other features of the EDA include support for randomization, blinding and power calculations, and procedures which are still underused in animal research[6]. Based on the diagram, the system generates a randomization sequence for the study, taking into

© 2017 Nature America, Inc., part of Springer Nature. All rights reserved.

nature research

Corresponding author(s): Nicola Segata and Levi Waldron

☐ Initial submission ☐ Revised version ☒ Final submission

Life Sciences Reporting Summary

Nature Research wishes to improve the reproducibility of the work that we publish. This form is intended for publication with all accepted life science papers and provides structure for consistency and transparency in reporting. Every life science submission will use this form; some list items might not apply to an individual manuscript, but all fields must be completed for clarity.

For further information on the points included in this form, see Reporting Life Sciences Research. For further information on Nature Research policies, including our data availability policy, see Authors & Referees and the Editorial Policy Checklist.

Experimental design

1. Sample size

Describe how sample size was determined.

curatedMetagenomicData is a database of human microbiome profiles from whole-metagenome shotgun sequencing from the Human Microbiome Project and as many other specimens collected from stool specimens as we have been able to find, curate, and process. Statistical parameters are not applicable for the database.

2. Data exclusions

Describe any data exclusions.

N/A

3. Replication

Describe whether the experimental findings were reliably reproduced.

All findings reported in the paper are reproducible using the script we provide at https://github.com/waldronlab/curatedMetagenomicData/blob/master/vignettes/extras/PaperFigures.Rmd

4. Randomization

Describe how samples/organisms/participants were allocated into experimental groups.

N/A

5. Blinding

Describe whether the investigators were blinded to group allocation during data collection and/or analysis.

N/A

Note: all studies involving animals and/or human research participants must disclose whether blinding and randomization were used.

6. Statistical parameters

For all figures and tables that use statistical methods, confirm that the following items are present in relevant figure legends (or in the Methods section if additional space is needed).

n/a	Confirmed	
☒	☐	The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement (animals, litters, cultures, etc.)
☐	☒	A description of how samples were collected, noting whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☒	☐	A statement indicating how many times each experiment was replicated
☒	☐	The statistical test(s) used and whether they are one- or two-sided (note: only common tests should be described solely by name; more complex techniques should be described in the Methods section)
☒	☐	A description of any assumptions or corrections, such as an adjustment for multiple comparisons
☒	☐	The test results (e.g. *P* values) given as exact values whenever possible and with confidence intervals noted
☒	☐	A clear description of statistics including central tendency (e.g. median, mean) and variation (e.g. standard deviation, interquartile range)
☒	☐	Clearly defined error bars

See the web collection on statistics for biologists for further resources and guidance.

Software

Policy information about availability of computer code

7. Software

Describe the software used to analyze the data in this study.	The offline pipeline for processing raw metagenomic data used publicly available MetaPhlAn2 and HUMAnN2 software. Code for all steps of downloading, proccessing, and generating the curatedMetagenomic package is available at www.github.com/waldronlab/curatedMetagenomicData, and the curatedMetagenomicData database is available through www.bioconductor.org.

For manuscripts utilizing custom algorithms or software that are central to the paper but not yet described in the published literature, software must be made available to editors and reviewers upon request. We strongly encourage code deposition in a community repository (e.g. GitHub). *Nature Methods* guidance for providing algorithms and software for publication provides further information on this topic.

Materials and reagents

Policy information about availability of materials

8. Materials availability

Indicate whether there are restrictions on availability of unique materials or if these materials are only available for distribution by a for-profit company.	N/A

9. Antibodies

Describe the antibodies used and how they were validated for use in the system under study (i.e. assay and species).	N/A

10. Eukaryotic cell lines

a. State the source of each eukaryotic cell line used.	N/A
b. Describe the method of cell line authentication used.	N/A
c. Report whether the cell lines were tested for mycoplasma contamination.	N/A
d. If any of the cell lines used are listed in the database of commonly misidentified cell lines maintained by ICLAC, provide a scientific rationale for their use.	N/A

Animals and human research participants

Policy information about studies involving animals; when reporting animal research, follow the ARRIVE guidelines

11. Description of research animals

Provide details on animals and/or animal-derived materials used in the study.	N/A

Policy information about studies involving human research participants

12. Description of human research participants

Describe the covariate-relevant population characteristics of the human research participants.	This study involves only use of publicly available data and does not qualify as Human Subject Research.